

AgriSmart AI

Crop disease detection from a single leaf photo — with weather risk, irrigation guidance, a sustainability score, and AI advice in the farmer's own language.

TEAM CURIOUS CODERS
MobileNetV3-Large · PlantVillage
macro-F1 0.9618 · acc 97.2%

01 · PROBLEM

Smallholder farmers detect crop disease late, and expert diagnosis is rarely reachable in time. Misdiagnosis leads to blanket pesticide use — cost, yield loss and soil damage at once.

A lightweight CNN classifier runs on a phone-quality leaf photo, then deterministic rule modules add weather, irrigation and sustainability context. A generative layer narrates only those computed numbers — it never diagnoses.

03 · SYSTEM ARCHITECTURE



04 · MODEL & TRAINING

BACKBONE	MobileNetV3-Large, ImageNet-pretrained
TRANSFER	Two-stage: frozen-backbone warm-up, then full fine-tune at 10× lower learning rate
DATA	PlantVillage — 54,305 lab-condition images, 38 classes, 14 crops
LOSS	Class-weighted cross-entropy — an 86-image class contributes as much gradient signal as an 800-image one
AUGMENT	Crop, flip, rotation, colour-jitter, blur, random erasing — simulating phone-camera noise
STOPPING	Early stopping on validation macro-F1
METRIC	Macro-F1, not accuracy — every class weighted equally regardless of sample count

05 · RESULTS

MEASURE	VALUE	NOTE
Validation accuracy	97.2%	held-out split
Validation macro-F1	0.9618	primary metric
Best epoch	24	early-stopped on macro-F1
Weakest class F1	0.784	Corn Cercospora, 86 samples

Corn Cercospora Gray Leaf Spot is confused mainly with two other corn diseases — Common Rust and Northern Leaf Blight — not with unrelated crops, so errors stay within a plausible treatment neighbourhood. Full confusion matrix overleaf.

06 · BONUS MODULES — deterministic rule-based logic, not additional ML models

WX	Weather risk — Open-Meteo forecast, threshold-based disease advisories.	SUS	Sustainability score — transparent 0–100, weighting formula shown beside the number.
IRR	Irrigation — soil-moisture plus rain-forecast rule returns water / hold.	GEN	AI explanation — Gemini narrates computed figures; instructed never to invent a diagnosis.
I18N	Six-language UI — English, Hindi, Gujarati, Marathi, Tamil, Telugu.		

07 · KNOWN LIMITATIONS

- **Domain gap.** Trained only on lab-condition images; a field photo can yield a confident but wrong prediction.
- **Corn confusion.** Gray Leaf Spot, Common Rust and Northern Leaf Blight overlap visually at low sample counts.
- **Single-leaf only.** Whole-image classification, not multi-leaf detection or localisation.
- **No API auth.** Firebase gates frontend pages only; Flask routes are currently unauthenticated.

Confusion matrix — 38 classes, validation split

Rows are true labels, columns predicted. The diagonal dominates; residual mass sits inside crop families — most visibly the three corn diseases — rather than across unrelated crops, which is the failure mode we would accept in the field.

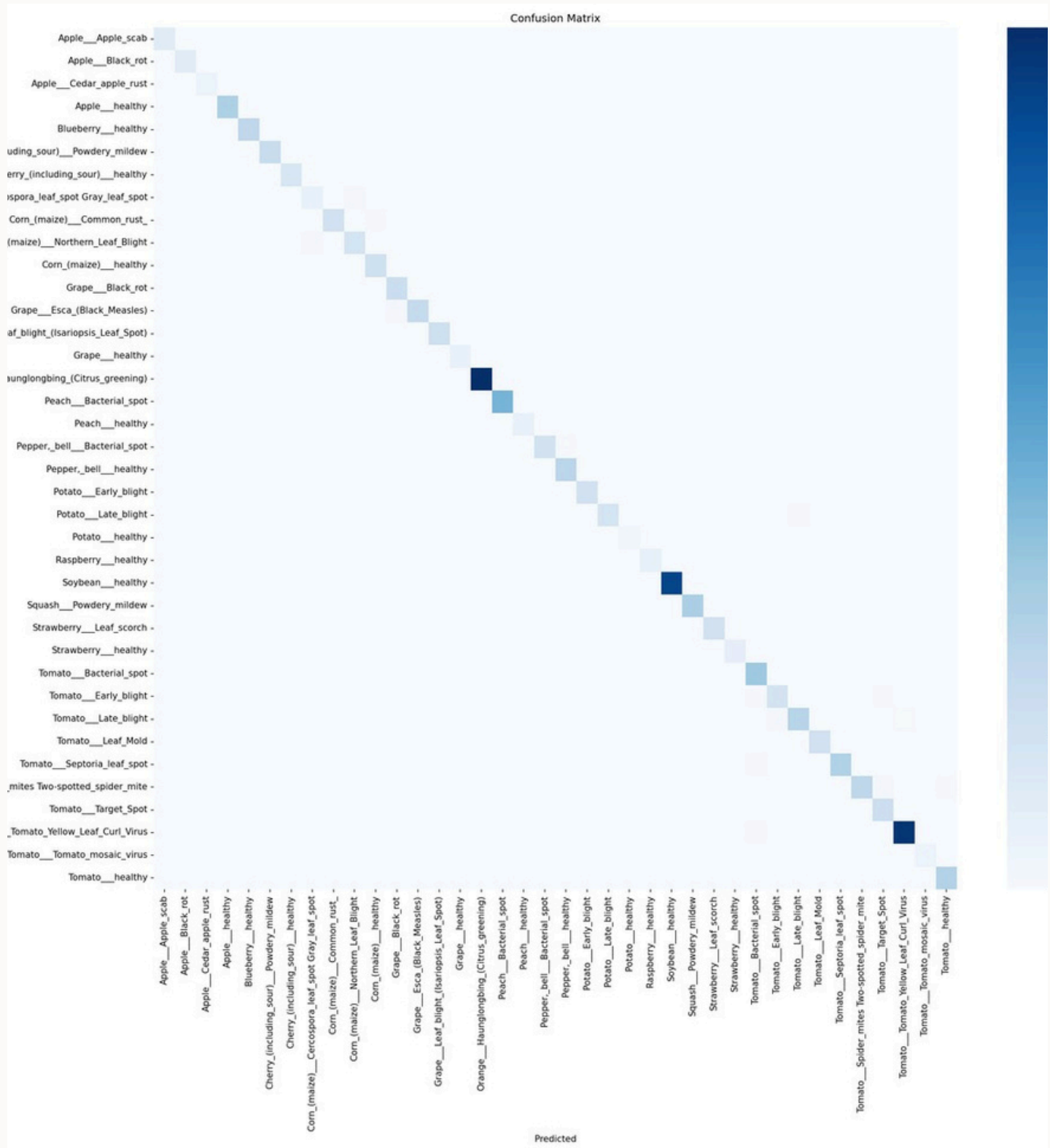


Figure A1. Row-normalised confusion matrix, best epoch 24 (validation accuracy 97.2%, macro-F1 0.9618). Weakest cell: Corn Cercospora Gray Leaf Spot, F1 0.784 on 86 training samples.